BZ/CL Spread Analysis — Grid Recalibration

Brent (BZUSDT) vs WTI (CLUSDT) perpetuals on Bybit · hourly closes, May 13 – Jul 6, 2026 (53 days, 1,294 bars)

MEAN SPREAD
3.94%
structural Brent premium

STD DEV
0.79%
hourly closes

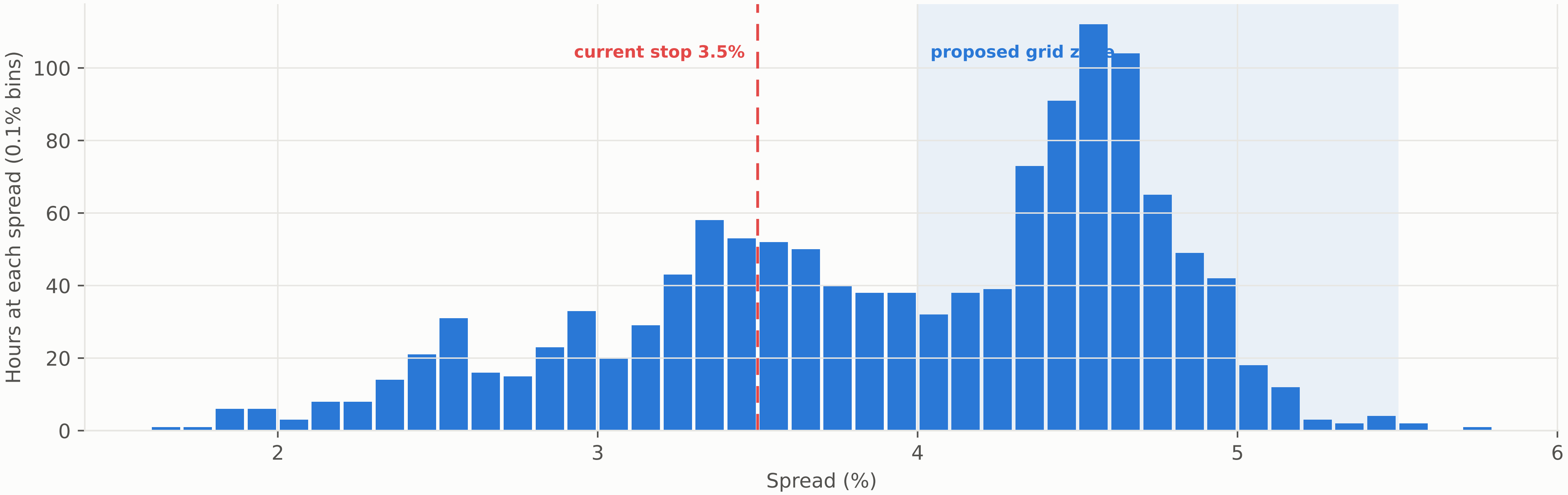
MIN
1.69%
never reaches current grid

MAX
5.71%
above current 3.5% stop

TIME ABOVE 3.5% STOP
69.9%
current config broken

Spread history vs current & proposed grid

Spread distribution



Where do closes actually happen?

A grid level only earns if the spread crosses it downward (open above → close below). Down-crossings over the 53-day window:

Level	Down-crossings	≈ per day	Time above
3.50%	31	0.58	69.9%
3.75%	29	0.55	60.2%
4.00%	27	0.51	53.0%
4.25%	27	0.51	46.3%
4.50%	**52**	**0.98**	**31.9%**
4.75%	25	0.47	12.2%
5.00%	16	0.30	3.2%
5.25%	4	0.08	0.8%

Most active reversion zone: 4.3–4.8%
(one close opportunity ≈ per day at 4.50%)

Recommendation

The current config cannot work for BZ/CL. It was inherited from PAXG/XAUT, where both legs track gold and the spread oscillates around 0. Brent carries a structural premium over WTI: all 14 current levels (0.10–3.00%) open immediately and can essentially never close (spread minimum was 1.69%), while extreme_spread_stop=3.5% sits below the mean — the strategy would be in permanent emergency stop.

Recenter the grid on the observed oscillation range:

```
grid_levels=[
    0.0400,  # 4.00% — open ~53% of the time
    0.0420,  # 4.20%
    0.0440,  # 4.40%
    0.0460,  # 4.60% — most active reversion zone
    0.0480,  # 4.80%
    0.0500,  # 5.00%
    0.0525,  # 5.25% — tail level
],
extreme_spread_stop=0.060,   # above observed max 5.71%
```

- Spacing 0.20–0.25%: the P90 hourly move is 0.24%, so adjacent levels aren't jumped constantly but still cycle roughly daily in the 4.3–4.8% zone.
- Weights: keep the light-at-bottom / heavy-at-top shape, but the "bottom" is now 4.00% (0.6× → 2.0× from 4.00% to 5.25%).
- Stop at 6.0%: clears the 53-day max (5.71%) while still capping tail blowout.

Caveats

- Short history: BZUSDT listed on Bybit only ~May 13, 2026 — 53 days is a thin sample. Treat these levels as provisional and re-run this analysis monthly.
- Upward drift: the monthly mean moved 3.85% → 3.88% → 4.64% (May → Jul). A static grid centered on a drifting mean goes stale; consider a rolling-mean-centered spread definition in the strategy as a follow-up change.
- Funding not modeled: holding short BZ / long CL across funding intervals adds a carry term this price-only analysis ignores.
- Convergence is not guaranteed: Brent–WTI is driven by real supply/demand (WTI is landlocked; Brent is seaborne). The premium can widen persistently — this is not a hard-pegged pair like PAXG/XAUT.